

AI Financial Document Q&A; Assistant

Development Backlog

Total tasks: 45 | Total story points: 123

Epic 0: Project Foundation

4 tasks, 7 points

#	Task	Priority	Points
1	Repo scaffolding (folder structure, requirements, pre-commit, README stub)	P0	2
2	AWS account setup: IAM user, S3 bucket, EC2 key pair	P0	2
3	Obtain LLM API keys (OpenAI/Gemini), store as env vars	P0	1
4	Base Dockerfile + docker-compose skeleton	P0	2

Epic 1: Ingestion Pipeline

7 tasks, 24 points

#	Task	Priority	Points
5	PDF upload handler + push raw file to S3	P0	3
6	Plain-text extraction for narrative sections	P0	3
7	Table-aware extraction for financial statements	P0	5
8	Chunking strategy (recursive splitter, intact table blocks)	P0	5
9	Metadata tagging on chunks (doc ID, page, section type)	P0	2
10	Embedding generation + upsert into ChromaDB	P0	3
11	Unit tests for chunking/table boundaries	P1	3

Epic 2: Retrieval Agent

4 tasks, 11 points

#	Task	Priority	Points
12	Query embedding + ChromaDB similarity search (top-k, k=5-8)	P0	3
13	Citation formatting (chunk to page/section)	P0	2
14	Basic single-agent Q&A; end-to-end	P0	3
15	Optional reranking step	P2	3

Epic 3: Multi-Agent Orchestration

9 tasks, 30 points

#	Task	Priority	Points
16	LangGraph state graph skeleton	P0	3
17	Intent router node	P0	3
18	Wire Retrieval Agent into graph	P0	2
19	KPI Extraction Agent: Pydantic schema	P0	5
20	KPI Extraction Agent: function-calling integration	P0	5
21	KPI cross-checking logic (repeated values)	P1	3
22	Summary Agent: map-reduce for full-report summaries	P0	5
23	Summary Agent: single-pass for section summaries	P1	2
24	State reuse of already-extracted KPIs on follow-ups	P1	2

Epic 4: Conversational Memory

3 tasks, 8 points

#	Task	Priority	Points
25	In-memory checkpointer for local dev	P0	2
26	Multi-turn follow-up handling	P0	3
27	SQLite-backed checkpointer for deployment	P1	3

Epic 5: Frontend (Streamlit)

4 tasks, 10 points

#	Task	Priority	Points
28	Upload panel wired to ingestion pipeline	P0	2
29	Chat panel wired to orchestrator	P0	3
30	KPI dashboard table	P0	3
31	UI polish (loading/error/empty states)	P1	2

Epic 6: Deployment & Infrastructure

5 tasks, 11 points

#	Task	Priority	Points
32	Finalize Dockerfile + docker-compose	P0	3
33	Provision EC2 instance	P0	2
34	Deploy + verify S3 connectivity	P0	3
35	CloudWatch logging	P1	2
36	S3 lifecycle rule for document expiry	P1	1

Epic 7: CI/CD Testing & Security

9 tasks, 22 points

#	Task	Priority	Points
37	GitHub Actions build on push to main	P1	3
38	SSH deploy step to EC2	P1	2
39	KPI schema validation unit tests	P0	2
40	Integration test: full ingest-query pipeline	P1	3
41	RAG evaluation set (10-15 Q/A pairs)	P1	3
42	S3 encryption + public-access block verification	P0	1
43	Password gate for public demo	P1	2
44	Rate limiter integration on LLM endpoints	P2	3
45	Test against 2-3 real-world annual reports	P1	3